



## Review Test Submission: Assignment Week 4

User Mr W (Wouter) W Nel

Course EBN 122 S2 2017

Test Assignment Week 4

Started 8/14/17 8:52 PM

Submitted 8/14/17 11:50 PM

Due Date 8/14/17 11:59 PM

Status Completed

Attempt Score 17 out of 20 points

Time Elapsed 2 hours, 58 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate  $\Omega$  (use sigular)
- Use 'u' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

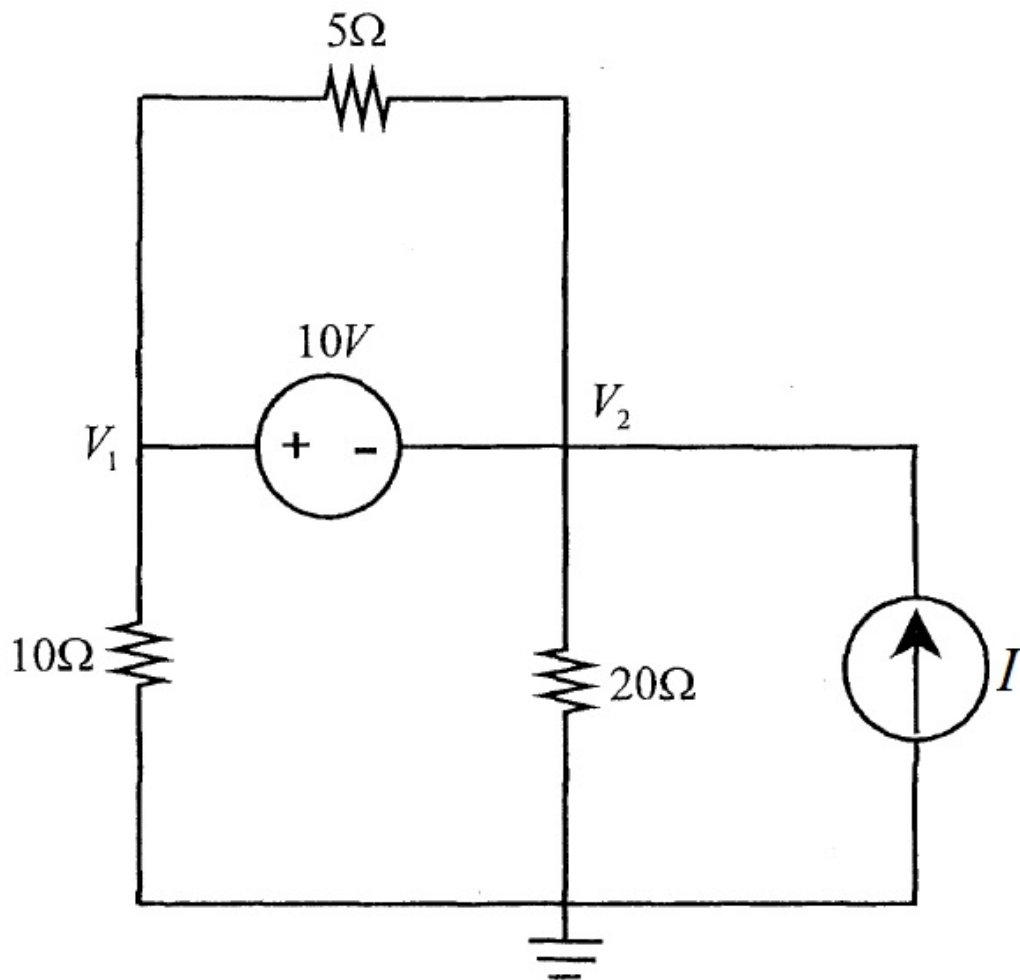
Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions Displayed

### Question 1

2 out of 2 points



Using nodal analysis, find  $V_1$  if  $I = 9$  A.



Selected Answer: ☒ 63.3V

Correct Answer: ☒ 63.33 ± 5% (V)

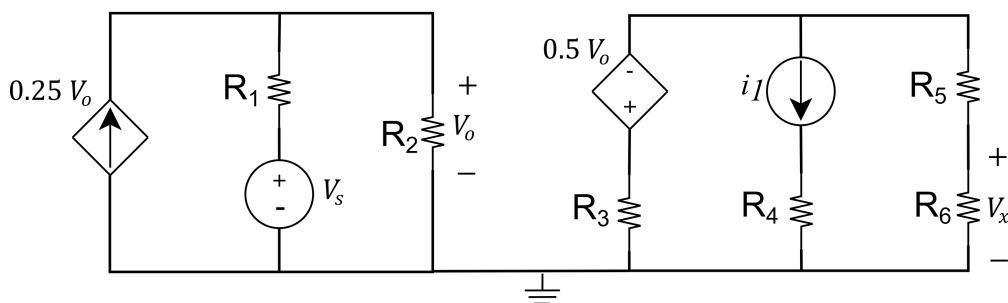
### Question 2

2 out of 2 points



Use nodal analysis to solve for  $V_x$ .

☒  $R_3=27\Omega$ ,  $R_4=56\Omega$ ,  $R_5=53\Omega$ ,  $R_6=47\Omega$ ,  $V_0=14V$ ,  $i_1=2A$

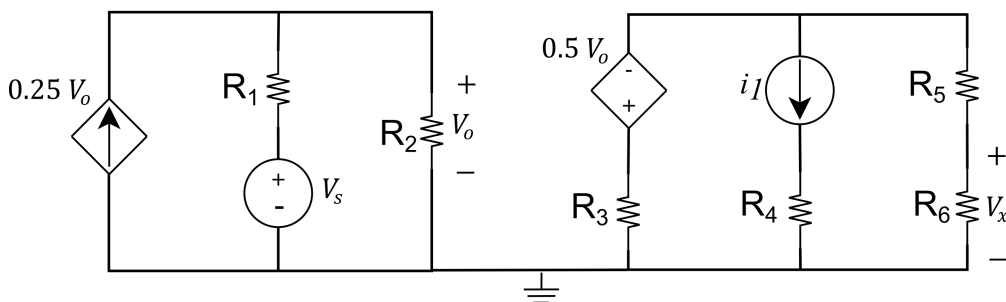


Selected Answer: ☒ -22.6V

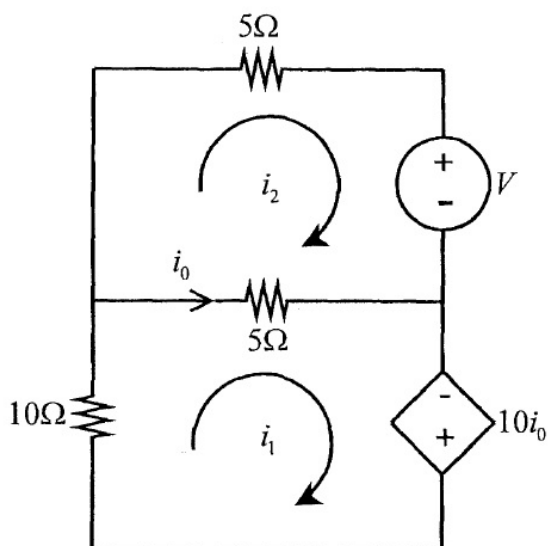
Correct Answer: ☒ -22.575 ± 2 (V)

### Question 3

2 out of 2 points

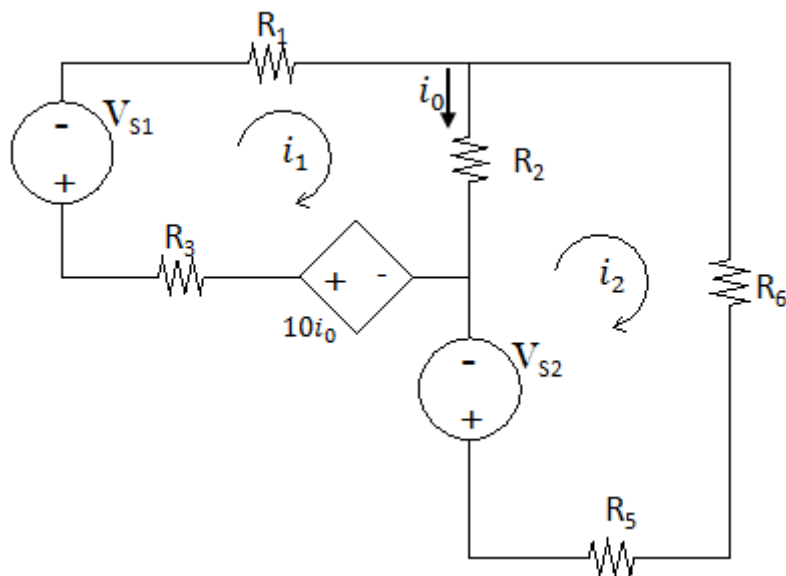
Use nodal analysis to solve for  $V_0$ .
 $R_1=13\Omega$ ,  $R_2=65\Omega$ ,  $R_3=92\Omega$ ,  $R_4=82\Omega$ ,  $R_5=35\Omega$ ,  $R_6=43\Omega$  and  $V_s=13V$ 
Selected Answer: ☒ -6.34VCorrect Answer: ☒ -6.341  $\pm$  5 (V)**Question 4**

0.5 out of 2 points

Using mesh analysis find  $i_1$  if  $V=21V$ Selected Answer: ☒ 2.8ACorrect Answer: ☒ 1.40  $\pm$  2% (A)**Question 5**

0.5 out of 2 points

Determine  $i_o$ .
 $R_1=62\Omega$ ,  $R_2=15\Omega$ ,  $R_3=70\Omega$ ,  $R_5=86\Omega$ ,  $R_6=95\Omega$ ,  $V_{S1}=13V$  and  $V_{S2}=6V$



Selected Answer: ✗ 58.3mA

Correct Answer: ✓ -58.296 ± 5% (mA)

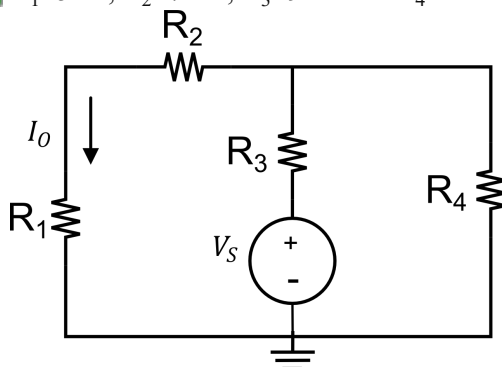
### Question 6

2 out of 2 points



If  $I_0 = 9\text{mA}$ , what is  $V_S$ ?

✓  $R_1=5\text{k}\Omega$ ,  $R_2=7\text{k}\Omega$ ,  $R_3=9\text{k}\Omega$  and  $R_4=4\text{k}\Omega$



Selected Answer: ✓ 432V

Correct Answer: ✓ 432.000 ± 5% (V)

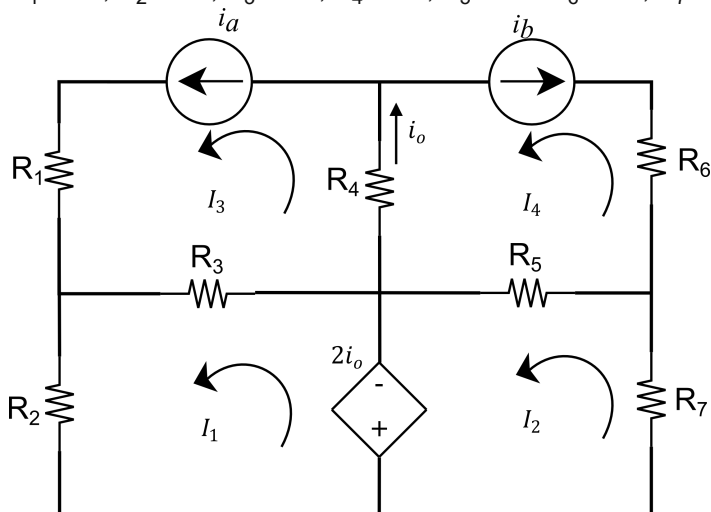
### Question 7

2 out of 2 points



Use mesh analysis to solve  $I_1$

$R_1=51\Omega$ ,  $R_2=44\Omega$ ,  $R_3=94\Omega$ ,  $R_4=66\Omega$ ,  $R_5=49\Omega$ ,  $R_6=37\Omega$ ,  $R_7=25\Omega$ ,  $i_a=3A$  and  $i_b=4A$



Selected Answer: ☒ 1.94A

Correct Answer: ☒ 1.942 ± 5% (A)

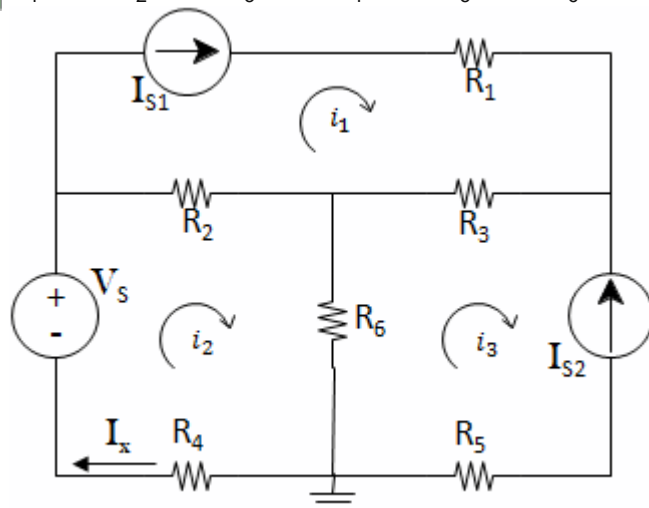
### Question 8

2 out of 2 points



Determine  $I_x$ , with

$R_1=42\Omega$ ,  $R_2=88\Omega$ ,  $R_3=34\Omega$ ,  $R_4=10\Omega$ ,  $R_5=14\Omega$ ,  $R_6=41\Omega$ ,  $I_{S1}=24mA$ ,  $I_{S2}=16mA$  and  $V_S=5V$



Selected Answer: ☒ 46.4mA

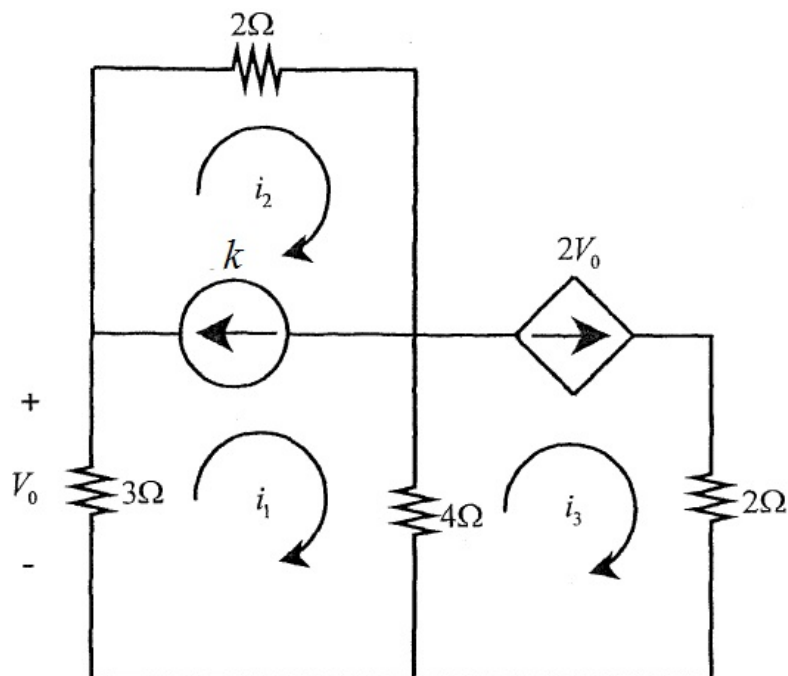
Correct Answer: ☒ 46.446 ± 5% (mA)

### Question 9

2 out of 2 points



If  $k = 29 A$ , determine  $i_2$



Selected Answer: ☒ 27.2A

Correct Answer: ☒  $27.24 \pm 2\%$  (A)

### Question 10

2 out of 2 points



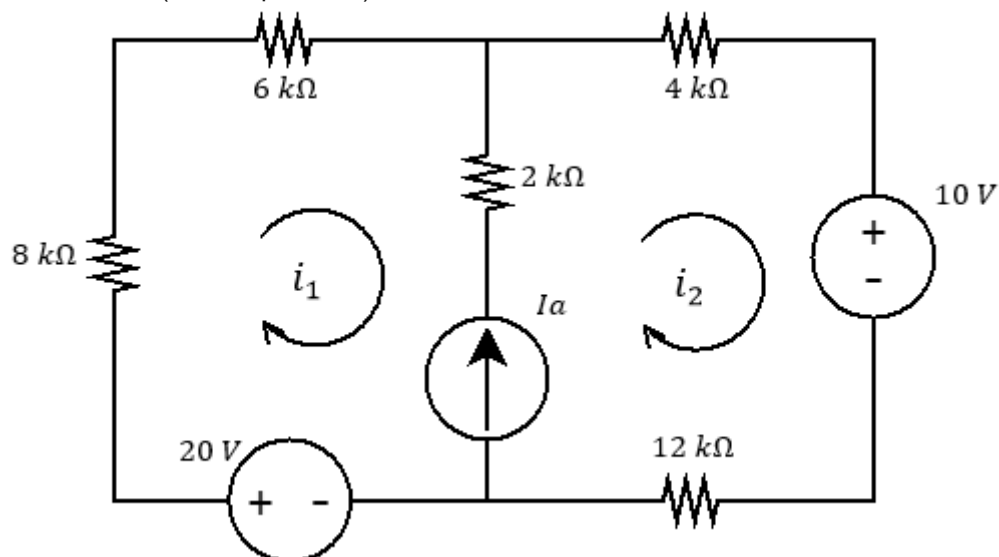
Use mesh analysis to set up two equations of the following form: Provide d, if  $I_a = 7.66$  mA.



(Answer needs to be in mA).

a  $i_1 + b i_2 = 10$  (Hint: KVL)

-  $i_1 + c i_2 = d$  (Hint: Supermesh)



Selected Answer: ☒ 7.66mA

Correct Answer: ☒  $7.66 \pm 5\%$

Monday, August 14, 2017 11:53:54 PM SAST

← OK